

scGen Human Validation — Reviewer 1 (Blinded)

Purpose

Independently determine how each citing paper uses scGen. Machine labels, confidence, agent outputs, and machine-selected evidence are intentionally hidden.

Workflow

Complete every row in the displayed order. Do not open the other reviewer document or the machine-selection file until both independent reviews are locked.

Use only the linked approved article versions. Inspect every scGen occurrence and its surrounding methods, results, figures, tables, and available supplements.

After both reviewers lock their answers, discuss only disagreements in a separate consensus record.

Allowed values

Review status: DRAFT / LOCKED

Citation validity and method executed: Yes / No / Unclear

Uses (select all): Mention only / Benchmark / Extend or develop / Apply to biological data / Other executed use / Unclear

Primary use: Mention only / Benchmark / Extend or develop / Apply to biological data / Other / Unclear

Biological insight: Yes / No / Unclear

Evidence sufficiency: Sufficient / Incomplete / Citation not located / Unclear

Supporting evidence: shortest exact quote plus section, page, paragraph, figure, or table locator.

Review table

Record	Article and approved evidence links	Reviewer 1 independent record
SCGEN-HV-022	Application and prospects of single-cell and spatial omics technologies in woody plants	Review status:LOCKED
FORMAL VALIDATION	DOI: 10.48130/fr-2023-0027 Article page: https://doi.org/10.48130/fr-2023-0027 Full text 1: https://www.maxapress.com/data/article/forres/preview/pdf/FR-2023-0027.pdf Controller-registered target occurrence locator(s): occ-0001 (document_4f75272e99730f259da1.txt:p-0008); occ-0002 (document_4f75272e99730f259da1.txt:p-0037)	Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: For comprehensive analyses and comparisons, it is often necessary to integrate data from different samples or experimental batches, which can be accomplished

		by software such as Harmony^[53], LIGER^[54], scMerge^[55], scGen^[56], and Seurat3^[48]. Notes:
SCGEN-HV-025 FORMAL VALIDATION	VI-VS: calibrated identification of feature dependencies in single-cell multiomics DOI: 10.1186/s13059-024-03419-z Article page: https://doi.org/10.1186/s13059-024-03419-z Full text 1: https://genomebiology.biomedcentral.com/counter/pdf/10.1186/s13059-024-03419-z Controller-registered target occurrence locator(s): occ-0001 (document_cfe0d58ca6c7605aef38.txt:p-0035)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: In this model, the prior is usually the standard normal, and the posterior distribution is approximated using a variational approach, with the approximation parameterized by neural networks [32, 74, 75]. Notes:
SCGEN-HV-001 FORMAL VALIDATION	Opportunities for Artificial Intelligence in Advancing Precision Medicine DOI: 10.1007/s40142-019-00177-4 Article page: https://doi.org/10.1007/s40142-019-00177-4 Full text 1: https://link.springer.com/content/pdf/10.1007/s40142-019-00177-4.pdf	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only

	Controller-registered target occurrence locator(s): occ-0001 (document_d89a75e950e99bb0d716.txt:p-0006)	Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: In addition, it is possible to predict cycle, disease progression, and perturbation responses using deep network approaches [41•, 42•, 43–45]. Notes:
SCGEN-HV-041 FORMAL VALIDATION	A Roadmap for a Consensus Human Skin Cell Atlas and Single-Cell Data Standardization DOI: 10.1016/j.jid.2023.03.1679 Article page: https://doi.org/10.1016/j.jid.2023.03.1679 Full text 1: https://uu.diva-portal.org/smash/get/diva2:1832264/FULLTEXT01 Controller-registered target occurrence locator(s): occ-0001 (document_340f03fd4110e8bb63f7.txt:p-0005); occ-0002 (document_340f03fd4110e8bb63f7.txt:p-0010)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Some methods, for example, fastMNN (Haghverdi et al., 2018), scANVI (Xu et al., 2021), and scGen (Lotfollahi et al., 2019), are more effective at ensuring biological conservation, at the risk of retaining some batch effects, whereas others, for example, BBKNN (Polański et al., 2020), Scanorama (Hie et al., 2019), and trVAE (Lotfollahi et al., 2020), are more effective at batch correction at the risk of obscuring biological variation (Luecken et al., 2022). Notes:

SCGEN-HV-045 FORMAL VALIDATION	scCompressSA: dual-channel self-attention based deep autoencoder model for single-cell clustering by compressing gene–gene interactions DOI: 10.1186/s12864-024-10286-2 Article page: https://doi.org/10.1186/s12864-024-10286-2 Full text 1: https://www.researchsquare.com/article/rs-3674838/latest.pdf Controller-registered target occurrence locator(s): occ-0001 (document_62463667ea5a15b4ae77.txt:p-0018)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Cell clustering of scRNA-seq data is crucial to detect meaningful patterns from raw gene count matrix [1,2,3]. Notes:
SCGEN-HV-040 FORMAL VALIDATION	Differences between in vivo and ex vivo models drive divergent perturbation outcomes DOI: 10.1101/2025.09.02.673692 Article page: https://doi.org/10.1101/2025.09.02.673692 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/09/07/2025.09.02.673692.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_9dd3e890037d378b8400.txt:p-0055)	Review status:LOCKED Citation validity:No Method executed:No Uses (select all): Primary use:Other Biological insight:No Evidence sufficiency:Insufficient Supporting quote + locator: Notes:
SCGEN-HV-035	lute: estimating the cell composition of heterogeneous tissue with varying cell sizes using gene expression DOI: 10.1186/s12864-025-11508-x Article page: https://doi.org/10.1186/s12864-025-11508-x	Review status:LOCKED Citation validity:Yes Method executed:No

FORMAL VALIDATION	Full text 1: https://www.biorxiv.org/content/biorxiv/early/2024/04/06/2024.04.04.588105.full.pdf Full text 2: https://bmcbgenomics.biomedcentral.com/counter/pdf/10.1186/s12864-025-11508-x Controller-registered target occurrence locator(s): occ-0001 (document_b94abc966b9c57ca4cb9.txt:p-0025); occ-0002 (document_ee70a2ea12d279c619be.txt:p-0019)	Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: However, each of these tools were built on different frameworks with non-uniform input data formats while addressing different types of systematic errors or unwanted bias [19,20,21,22,23,24,25,26]. Notes:
SCGEN-HV-010 FORMAL VALIDATION	Learning Perturbation-specific Cell Representations for Prediction of Transcriptional Response across Cellular Contexts DOI: 10.1101/2023.03.20.533433 Article page: https://doi.org/10.1101/2023.03.20.533433 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/03/23/2023.03.20.533433.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_01301fc720e62a99efc8.txt:p-0002); occ-0002 (document_01301fc720e62a99efc8.txt:p-0003); occ-0003 (document_01301fc720e62a99efc8.txt:p-0008); occ-0004 (document_01301fc720e62a99efc8.txt:p-0008); occ-0005 (document_01301fc720e62a99efc8.txt:p-0009); occ-0006 (document_01301fc720e62a99efc8.txt:p-0009); occ-0007 (document_01301fc720e62a99efc8.txt:p-0009); occ-0008 (document_01301fc720e62a99efc8.txt:p-0009); occ-0009 (document_01301fc720e62a99efc8.txt:p-0011); occ-0010 (document_01301fc720e62a99efc8.txt:p-0011); occ-0011 (document_01301fc720e62a99efc8.txt:p-0011); occ-0012 (document_01301fc720e62a99efc8.txt:p-0011); occ-0013 (document_01301fc720e62a99efc8.txt:p-0013)	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: We benchmarked the predictive performance of PerturbX to 1) a baseline model that discards cell line information and predicts the average effect (DE) over all the training cell lines, and 2) scGen (Lotfollahi et al., 2019), which utilizes a conditional VAE and latent space arithmetics to predict the perturbed single cell

		distribution in unseen cell types. Notes:
SCGEN-HV-002 FORMAL VALIDATION	Species-agnostic transfer learning for cross-species transcriptomics data integration without gene orthology DOI: 10.1093/bib/bbae004 Article page: https://doi.org/10.1093/bib/bbae004 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/08/14/2023.08.11.552752.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_79d3810f985d5821fe1d.txt:p-0003); occ-0002 (document_79d3810f985d5821fe1d.txt:p-0005); occ-0003 (document_79d3810f985d5821fe1d.txt:p-0005); occ-0004 (document_79d3810f985d5821fe1d.txt:p-0034)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Previously, scGEN has shown that vector arithmetics on latent space can predict gene signatures of unseen cell types in different species in the lps-stimulated macrophage dataset [35]. Likewise, we analyzed SATL on the same dataset. Notes:
SCGEN-HV-021 FORMAL VALIDATION	Combinatorial prediction of therapeutic perturbations using causally inspired neural networks DOI: 10.1038/s41551-025-01481-x Article page: https://doi.org/10.1038/s41551-025-01481-x Full text 1: https://www.nature.com/articles/s41551-025-01481-x.pdf Controller-registered target occurrence locator(s): occ-0001 (document_6fd74ea9f2adc84c8130.txt:p-0003); occ-0002 (document_6fd74ea9f2adc84c8130.txt:p-0005); occ-0003 (document_6fd74ea9f2adc84c8130.txt:p-0006); occ-0004 (document_6fd74ea9f2adc84c8130.txt:p-0007); occ-0005 (document_6fd74ea9f2adc84c8130.txt:p-0008); occ-0006	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: PDGrapher achieves the

	(document_6fd74ea9f2adc84c8130.txt:p-0008); occ-0007 (document_6fd74ea9f2adc84c8130.txt:p-0008); occ-0008 (document_6fd74ea9f2adc84c8130.txt:p-0009); occ-0009 (document_6fd74ea9f2adc84c8130.txt:p-0015); occ-0010 (document_6fd74ea9f2adc84c8130.txt:p-0015); occ-0011 (document_6fd74ea9f2adc84c8130.txt:p-0015); occ-0012 (document_6fd74ea9f2adc84c8130.txt:p-0015); occ-0013 (document_6fd74ea9f2adc84c8130.txt:p-0015); occ-0014 (document_6fd74ea9f2adc84c8130.txt:p-0015); occ-0015 (document_6fd74ea9f2adc84c8130.txt:p-0015); occ-0016 (document_6fd74ea9f2adc84c8130.txt:p-0015); occ-0017 (document_6fd74ea9f2adc84c8130.txt:p-0016); occ-0018 (document_6fd74ea9f2adc84c8130.txt:p-0017); occ-0019 (document_6fd74ea9f2adc84c8130.txt:p-0020); occ-0020 (document_6fd74ea9f2adc84c8130.txt:p-0020); occ-0021 (document_6fd74ea9f2adc84c8130.txt:p-0020); occ-0022 (document_6fd74ea9f2adc84c8130.txt:p-0021); occ-0023 (document_6fd74ea9f2adc84c8130.txt:p-0021); occ-0024 (document_6fd74ea9f2adc84c8130.txt:p-0021); occ-0025 (document_6fd74ea9f2adc84c8130.txt:p-0021); occ-0026 (document_6fd74ea9f2adc84c8130.txt:p-0021); occ-0027 (document_6fd74ea9f2adc84c8130.txt:p-0021); occ-0028 (document_6fd74ea9f2adc84c8130.txt:p-0021); occ-0029 (document_6fd74ea9f2adc84c8130.txt:p-0021)	second-best overall rankings (Extended Data Fig. 2d), closely following scGEN, which obtained the highest nDCG values (Extended Data Fig. 2b) but showed weaker performance when evaluating only the top-ranked predicted targets (partially accurate prediction and recall@1). Notes:
SCGEN-HV-046 FORMAL VALIDATION	Biological network-inspired interpretable variational autoencoder DOI: 10.1101/2020.12.17.423310 Article page: https://doi.org/10.1101/2020.12.17.423310 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2020/12/19/2020.12.17.423310.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_62ca3a461f8c2418ce78.txt:p-0008); occ-0002 (document_62ca3a461f8c2418ce78.txt:p-0008); occ-0003 (document_62ca3a461f8c2418ce78.txt:p-0025); occ-0004 (document_62ca3a461f8c2418ce78.txt:p-0033); occ-0005 (document_62ca3a461f8c2418ce78.txt:p-0051); occ-0006 (document_62ca3a461f8c2418ce78.txt:p-0080)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: For example, while theoretically our model could be used for interpretable response prediction using latent vector arithmetics in a similar fashion to scGen10, the limited generative capacity of our approach sacrifices predictive performance for interpretability Notes:

		Notes:
SCGEN-HV-042 FORMAL VALIDATION	Embeddings from language models are good learners for single-cell data analysis DOI: 10.1016/j.patter.2025.101431 Article page: https://doi.org/10.1016/j.patter.2025.101431 Full text 1: https://www.researchsquare.com/article/rs-4014784/latest.pdf Controller-registered target occurrence locator(s): occ-0001 (document_9023646e3bd30df62acb.txt:p-0025)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:Yes Evidence sufficiency:Sufficient Supporting quote + locator: To this end, we focused on three distinct tasks spanning both chemical perturbations¹⁰⁴ and gene-level perturbations (e.g., CRISPR-based perturb-seq). Notes:
SCGEN-HV-027 FORMAL VALIDATION	Single-cell RNA-seq reveals spatially restricted multicellular fibrotic niches during lung fibrosis DOI: 10.1101/569855 Article page: https://doi.org/10.1101/569855 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2019/03/06/569855.full.pdf No controller-registered target occurrence; independently audit citation location and evidence coverage.	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Computational consolidation of single-cell RNA sequencing datasets generated in other models of fibrosis is a promising approach to generate hypotheses with respect to fibrosis pathogenesis that can be tested with causal genetic or pharmacological interventions [39].

		Notes:
SCGEN-HV-033 FORMAL VALIDATION	Learning interpretable cellular and gene signature embeddings from single-cell transcriptomic data DOI: 10.1038/s41467-021-25534-2 Article page: https://doi.org/10.1038/s41467-021-25534-2 Full text 1: https://www.nature.com/articles/s41467-021-25534-2.pdf Full text 2: https://www.biorxiv.org/content/biorxiv/early/2021/06/10/2021.01.13.426593.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_792fe740c44cb4cb3ffb.txt:p-0002); occ-0002 (document_792fe740c44cb4cb3ffb.txt:p-0014); occ-0003 (document_1726b16a80b9449cb429.txt:p-0003); occ-0004 (document_1726b16a80b9449cb429.txt:p-0022)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Lotfollahi et al.²⁶ developed a VAE model called scGen to infer the expression difference due to perturbation conditions by latent space interpolation. Notes:
SCGEN-HV-034 FORMAL VALIDATION	Detecting subtle transcriptomic perturbations induced by lncRNAs knock-down in single-cell CRISPRi screening using a new sparse supervised autoencoder neural network DOI: 10.3389/fbinf.2024.1340339 Article page: https://doi.org/10.3389/fbinf.2024.1340339 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/07/11/2023.07.11.548494.full.pdf Full text 2: https://hal.science/hal-04795816v1/file/Truchi_Frontiers_Bioinfo24.pdf Controller-registered target occurrence locator(s): occ-0001 (document_d9f816edbc1045f84712.txt:p-0018)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: For example, VAEs have been applied to scRNA-seq to predict cell response to biological perturbations (Lotfollahi et al., 2019) and Omics dataset (Eltager et al., 2023).

		Notes:
SCGEN-HV-012 FORMAL VALIDATION	scVAE: variational auto-encoders for single-cell gene expression data. DOI: 10.1093/bioinformatics/btaa293 Article page: https://doi.org/10.1093/bioinformatics/btaa293 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2019/10/02/318295.full.pdf No controller-registered target occurrence; independently audit citation location and evidence coverage.	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: As mentioned above, it should also be possible to combine encodings of the cells in the latent space and produce in-between cells like Lotfollahi et al. (2018). Notes:
SCGEN-HV-037 FORMAL VALIDATION	Variational autoencoders learn universal latent representations of metabolomics data DOI: 10.1101/2021.01.14.426721 Article page: https://doi.org/10.1101/2021.01.14.426721 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2021/02/18/2021.01.14.426721.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_cefac28e16cb32969c7c.txt:p-0054)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: The VAE architecture has, for example, proven effective for predicting cell-level response to infection in transcriptomic data not available during training, and predicting drug response from gene expression data where drug

		response information is sparse¹⁹⁻²². Notes:
SCGEN-HV-005 FORMAL VALIDAT ION	Benchmarking LLM-based agents for single-cell omics analysis DOI: 10.1186/s13059-026-03998-z Article page: https://doi.org/10.1186/s13059-026-03998-z Full text 1: https://arxiv.org/pdf/2508.13201 Controller-registered target occurrence locator(s): occ-0001 (document_3ed8b7d7d2576fa8e251.txt:p-0020); occ-0002 (document_3ed8b7d7d2576fa8e251.txt:p-0020); occ-0003 (document_3ed8b7d7d2576fa8e251.txt:p-0022); occ-0004 (document_3ed8b7d7d2576fa8e251.txt:p-0063); occ-0005 (document_3ed8b7d7d2576fa8e251.txt:p-0064); occ-0006 (document_3ed8b7d7d2576fa8e251.txt:p-0082); occ-0007 (document_3ed8b7d7d2576fa8e251.txt:p-0100); occ-0008 (document_3ed8b7d7d2576fa8e251.txt:p-0100); occ-0009 (document_3ed8b7d7d2576fa8e251.txt:p-0111); occ-0010 (document_3ed8b7d7d2576fa8e251.txt:p-0111)	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: More correlation analysis for cell annotation (CellAssign), perturbation prediction (scGEN), gene imputation (SpaGE) and HVGs selection (Triku) are shown in Additional file 1: Fig. S7b. Notes:
SCGEN-HV-019 FORMAL VALIDAT ION	Shortcomings of silhouette in single-cell integration benchmarking DOI: 10.1038/s41587-025-02743-4 Article page: https://doi.org/10.1038/s41587-025-02743-4 Full text 1: https://www.nature.com/articles/s41587-025-02743-4.pdf Controller-registered target occurrence locator(s): occ-0001 (document_84897ab6cd25226ee536.txt:p-0006)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient

		Supporting quote + locator: From 2017 onward, silhouette-based metrics have also been used for scoring batch effect removal^{5,7,8,9}. Notes:
SCGEN-HV-049 FORMAL VALIDATION	Synthetic single cell RNA sequencing data from small pilot studies using deep generative models DOI: 10.1038/s41598-021-88875-4 Article page: https://doi.org/10.1038/s41598-021-88875-4 Full text 1: https://www.nature.com/articles/s41598-021-88875-4.pdf Controller-registered target occurrence locator(s): occ-0001 (document_d6332622f79e7f2a29c0.txt:p-0010)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Moreover, synthetic data can also be valuable for experimental design by investigating in-silico generated cell perturbations¹⁰. Notes:
SCGEN-HV-009 FORMAL VALIDATION	PerTurboAgent: A Self-Planning Agent for Boosting Sequential Perturb-seq Experiments DOI: 10.1101/2025.05.25.656020 Article page: https://doi.org/10.1101/2025.05.25.656020 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/05/27/2025.05.25.656020.full.pdf	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient

	Controller-registered target occurrence locator(s): occ-0001 (document_902e1d6b1e395793559e.txt:p-0013)	Supporting quote + locator: Indeed, knowledge about correlated gene expression patterns and shared gene functions has been leveraged for machine learning models, including generative and foundation models which aim to predict the impact of unseen perturbations (Roohani et al., 2024a; Bunne et al., 2023; Cui et al., 2024; Lotfollahi et al., 2019; 2023; Hao et al., 2024). Notes:
SCGEN-HV-004 FORMAL VALIDATION	scDREAMER for atlas-level integration of single-cell datasets using deep generative model paired with adversarial classifier DOI: 10.1038/s41467-023-43590-8 Article page: https://doi.org/10.1038/s41467-023-43590-8 Full text 1: https://www.nature.com/articles/s41467-023-43590-8.pdf Controller-registered target occurrence locator(s): occ-0001 (document_ee8eb663889e8c2359cf.txt:p-0008); occ-0002 (document_ee8eb663889e8c2359cf.txt:p-0008); occ-0003 (document_ee8eb663889e8c2359cf.txt:p-0008); occ-0004 (document_ee8eb663889e8c2359cf.txt:p-0008); occ-0005 (document_ee8eb663889e8c2359cf.txt:p-0008); occ-0006 (document_ee8eb663889e8c2359cf.txt:p-0010); occ-0007 (document_ee8eb663889e8c2359cf.txt:p-0010); occ-0008 (document_ee8eb663889e8c2359cf.txt:p-0010); occ-0009 (document_ee8eb663889e8c2359cf.txt:p-0010); occ-0010 (document_ee8eb663889e8c2359cf.txt:p-0010); occ-0011 (document_ee8eb663889e8c2359cf.txt:p-0010); occ-0012 (document_ee8eb663889e8c2359cf.txt:p-0010); occ-0013 (document_ee8eb663889e8c2359cf.txt:p-0010); occ-0014 (document_ee8eb663889e8c2359cf.txt:p-0011); occ-0015 (document_ee8eb663889e8c2359cf.txt:p-0011); occ-0016 (document_ee8eb663889e8c2359cf.txt:p-0012); occ-0017 (document_ee8eb663889e8c2359cf.txt:p-0012); occ-0018 (document_ee8eb663889e8c2359cf.txt:p-0012); occ-0019 (document_ee8eb663889e8c2359cf.txt:p-0012); occ-0020 (document_ee8eb663889e8c2359cf.txt:p-0012); occ-0021 (document_ee8eb663889e8c2359cf.txt:p-0014); occ-0022 (document_ee8eb663889e8c2359cf.txt:p-0014); occ-0023 (document_ee8eb663889e8c2359cf.txt:p-0018)	Review status: LOCKED Citation validity: Yes Method executed: Yes Uses (select all): Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: We compared scDREAMER-Sup's performance against that of scANVI³⁰ and scGEN³¹, state-of-the-art supervised methods (also top performers in recent benchmarking²⁴) that utilize cell type labels. Notes:
SCGEN-HV-043 FORMAL VALIDATION	Technical and legal aspects of federated learning in bioinformatics: applications, challenges and opportunities DOI: 10.3389/fdgth.2025.1644291 Article page: https://doi.org/10.3389/fdgth.2025.1644291	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all):

	Full text 1: https://public-pages-files-2025.frontiersin.org/journals/digital-health/articles/10.3389/fdgth.2025.1644291/pdf Controller-registered target occurrence locator(s): occ-0001 (document_0164d35eae5262e19dc3.txt:p-0009); occ-0002 (document_0164d35eae5262e19dc3.txt:p-0009); occ-0003 (document_0164d35eae5262e19dc3.txt:p-0009); occ-0004 (document_0164d35eae5262e19dc3.txt:p-0016); occ-0005 (document_0164d35eae5262e19dc3.txt:p-0016)	Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: More recently, Bakhtiari et al. (93) introduced FedscGen, a federated implementation of scGen (94), a variational autoencoder-based method for batch effect correction. Notes:
SCGEN-HV-014 FORMAL VALIDATION	Fluctuation structure predicts genome-wide perturbation outcomes DOI: 10.1101/2025.06.27.661814 Article page: https://doi.org/10.1101/2025.06.27.661814 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2025/07/01/2025.06.27.661814.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_8e25f50063871c9d0b5c.txt:p-0007); occ-0002 (document_8e25f50063871c9d0b5c.txt:p-0082)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Predictive models such as scGen²⁰, GEARS²¹, and scGPT²², on the other hand, employ deep learning frameworks to model perturbation responses in latent spaces and can predict outcomes of unseen perturbations. Notes:
SCGEN-HV-036 FORMAL VALIDATION	Learning cell communication from spatial graphs of cells DOI: 10.1101/2021.07.11.451750 Article page: https://doi.org/10.1101/2021.07.11.451750 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2021/07/12/2021.07.11.451750.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_653b2dec9c0512787d60.txt:p-0047)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient

		Supporting quote + locator: This node-generation tasks of learning a distribution over gene expression states conditioned on the cell-type, a local context in the graph, and other covariates, such as chemical perturbation, extends generative models for single-cell gene expression data^{21,23} to spatial cell-cell dependencies. Notes:
SCGEN-HV-024 FORMAL VALIDATION	Assessing and mitigating batch effects in large-scale omics studies DOI: 10.1186/s13059-024-03401-9 Article page: https://doi.org/10.1186/s13059-024-03401-9 Full text 1: https://link.springer.com/content/pdf/10.1186/s13059-024-03401-9.pdf Controller-registered target occurrence locator(s): occ-0001 (document_d011dcd0741d704029f6.txt:p-0028); occ-0002 (document_d011dcd0741d704029f6.txt:p-0040)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: DL methods usually use neural network algorithms to identify and remove batch effects from the data, including AutoClass [141], DESC [142], scGen [143], scVI [144], and so on. Notes:

SCGEN-HV-006 FORMAL VALIDATION	BuDDI: Bulk Deconvolution with Domain Invariance to predict cell-type-specific perturbations from bulk DOI: 10.1371/journal.pcbi.1012742 Article page: https://doi.org/10.1371/journal.pcbi.1012742 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/10/05/2023.07.20.549951.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_cff75af3fe5e992064e.txt:p-0015)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: To accomplish this goal, we build on ideas from three methodological approaches: bulk deconvolution [4–14], variational autoencoder (VAE) [15] models for perturbation prediction [16–22,23], and disentanglement methods [18,24–28]. Notes:
SCGEN-HV-029 FORMAL VALIDATION	Style transfer with variational autoencoders is a promising approach to RNA-Seq data harmonization and analysis DOI: 10.1093/bioinformatics/btaa624 Article page: https://doi.org/10.1093/bioinformatics/btaa624 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2020/04/28/791962.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_e4c1ddba928d429c11fb.txt:p-0006); occ-0002 (document_e4c1ddba928d429c11fb.txt:p-0006); occ-0003 (document_e4c1ddba928d429c11fb.txt:p-0006); occ-0004 (document_e4c1ddba928d429c11fb.txt:p-0006); occ-0005 (document_e4c1ddba928d429c11fb.txt:p-0010); occ-0006 (document_e4c1ddba928d429c11fb.txt:p-0010); occ-0007 (document_e4c1ddba928d429c11fb.txt:p-0018); occ-0008 (document_e4c1ddba928d429c11fb.txt:p-0031)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: The data were taken from scGen examples (https://github.com/theislabs/scgen-reproducibility). Notes:

SCGEN-HV-039 FORMAL VALIDATION	A deep generative model for deciphering cellular dynamics and in silico drug discovery in complex diseases DOI: 10.1038/s41551-025-01423-7 Article page: https://doi.org/10.1038/s41551-025-01423-7 Full text 1: https://digitalcommons.library.tmc.edu/cgi/viewcontent.cgi?article=7661&context=baylor_docs Controller-registered target occurrence locator(s): occ-0001 (document_b0f410bbf326127d9bce.txt:p-0006); occ-0002 (document_b0f410bbf326127d9bce.txt:p-0014); occ-0003 (document_b0f410bbf326127d9bce.txt:p-0016); occ-0004 (document_b0f410bbf326127d9bce.txt:p-0016); occ-0005 (document_b0f410bbf326127d9bce.txt:p-0016); occ-0006 (document_b0f410bbf326127d9bce.txt:p-0016); occ-0007 (document_b0f410bbf326127d9bce.txt:p-0016); occ-0008 (document_b0f410bbf326127d9bce.txt:p-0016); occ-0009 (document_b0f410bbf326127d9bce.txt:p-0017); occ-0010 (document_b0f410bbf326127d9bce.txt:p-0017); occ-0011 (document_b0f410bbf326127d9bce.txt:p-0017); occ-0012 (document_b0f410bbf326127d9bce.txt:p-0017); occ-0013 (document_b0f410bbf326127d9bce.txt:p-0017); occ-0014 (document_b0f410bbf326127d9bce.txt:p-0017); occ-0015 (document_b0f410bbf326127d9bce.txt:p-0023); occ-0016 (document_b0f410bbf326127d9bce.txt:p-0023); occ-0017 (document_b0f410bbf326127d9bce.txt:p-0023); occ-0018 (document_b0f410bbf326127d9bce.txt:p-0023); occ-0019 (document_b0f410bbf326127d9bce.txt:p-0023); occ-0020 (document_b0f410bbf326127d9bce.txt:p-0024); occ-0021 (document_b0f410bbf326127d9bce.txt:p-0024); occ-0022 (document_b0f410bbf326127d9bce.txt:p-0024); occ-0023 (document_b0f410bbf326127d9bce.txt:p-0025); occ-0024 (document_b0f410bbf326127d9bce.txt:p-0034)	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: To demonstrate UNAGI's advantages over existing methods in understanding the dynamics of diseases, we benchmarked it against established methods, including scVI¹⁹, GraphSCC²², scGEN³³, scGGAN²⁰, scGPT³¹, Geneformer¹⁴⁴, scGNN²¹, Seurat¹⁶ and SCANPY¹⁷, on the IPF dataset and scRNA COVID-19 PBMC data¹²⁸. Notes:
SCGEN-HV-044 FORMAL VALIDATION	Contrastive Mixture of Posteriors for Counterfactual Inference, Data Integration and Fairness DOI: 10.48550/arxiv.2106.08161 Article page: https://doi.org/10.48550/arxiv.2106.08161 Full text 1: https://arxiv.org/pdf/2106.08161 Controller-registered target occurrence locator(s): occ-0001 (document_9ba19ba18f3ae92f6ff2.txt:p-0008); occ-0002 (document_9ba19ba18f3ae92f6ff2.txt:p-0008); occ-0003 (document_9ba19ba18f3ae92f6ff2.txt:p-0011)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Predicting cell-level response to intervention has been previously tackled using VAEs by scGen

		(Lotfollahi et al., 2019b) and trVAE (Lotfollahi et al., 2019a). Notes:
SCGEN-HV-048 FORMAL VALIDATION	D-SPIN constructs gene regulatory network models from multiplexed scRNA-seq data revealing organizing principles of cellular perturbation response DOI: 10.1101/2023.04.19.537364 Article page: https://doi.org/10.1101/2023.04.19.537364 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/05/20/2023.04.19.537364.full.pdf No controller-registered target occurrence; independently audit citation location and evidence coverage.	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: The inclusion of higher-order multi-body interactions or concentration-dependent regulatory responses would enhance accuracy and predictive capabilities. ^{116,117,118,119} Notes:
SCGEN-HV-031 FORMAL VALIDATION	Learning genetic perturbation effects with variational causal inference DOI: 10.1371/journal.pcbi.1013194 Article page: https://doi.org/10.1371/journal.pcbi.1013194 Full text 1: https://journals.plos.org/ploscompbiol/article/file?id=10.1371/journal.pcbi.1013194&type=printable Controller-registered target occurrence locator(s): occ-0001 (document_0d4ecf1c37b93e695e22.txt:p-0017)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Prior work have also used Gaussian-based latent modeling of single-cell data [30,31].

	(document_469515c40259229a09e3.txt:p-0007); occ-0012 (document_469515c40259229a09e3.txt:p-0011); occ-0013 (document_469515c40259229a09e3.txt:p-0013); occ-0014 (document_469515c40259229a09e3.txt:p-0013); occ-0015 (document_469515c40259229a09e3.txt:p-0013); occ-0016 (document_469515c40259229a09e3.txt:p-0013); occ-0017 (document_469515c40259229a09e3.txt:p-0013); occ-0018 (document_469515c40259229a09e3.txt:p-0013); occ-0019 (document_469515c40259229a09e3.txt:p-0014); occ-0020 (document_469515c40259229a09e3.txt:p-0014); occ-0021 (document_469515c40259229a09e3.txt:p-0014); occ-0022 (document_469515c40259229a09e3.txt:p-0017); occ-0023 (document_469515c40259229a09e3.txt:p-0018)	CoCoA-diff³⁷, scGen²², CPA²³, ContrastiveVI²⁴, and CellOT²⁵, with two variants of our approach with or without sampling (CINEMA-OT-W, CINEMA-OT). Notes:
SCGEN-HV-011 FORMAL VALIDATION	A Cross-Modal Autoencoder Framework Learns Holistic Representations of Cardiovascular State DOI: 10.1101/2022.05.26.493497 Article page: https://doi.org/10.1101/2022.05.26.493497 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2022/05/28/2022.05.26.493497.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_15f5b7d997bf75c8406b.txt:p-0011)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: These models have been successfully applied in a variety of applications including computer vision^{12,13,14}, chemistry¹⁵, and biology^{16,17,18,19,20,21}. Notes:
SCGEN-HV-047 FORMAL VALIDATION	Clifti-GPT: privacy-preserving federated fine-tuning and transferable inference of foundation models on clinical single-cell data DOI: 10.1186/s13040-026-00582-w Article page: https://doi.org/10.1186/s13040-026-00582-w Full text 1: https://link.springer.com/content/pdf/10.1186/s13040-026-00582-w.pdf	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use:Apply to biological data

	Controller-registered target occurrence locator(s): occ-0001 (document_230f106da00218449171.txt:p-0017); occ-0002 (document_230f106da00218449171.txt:p-0017); occ-0003 (document_230f106da00218449171.txt:p-0025); occ-0004 (document_230f106da00218449171.txt:p-0030); occ-0005 (document_230f106da00218449171.txt:p-0030); occ-0006 (document_230f106da00218449171.txt:p-0036); occ-0007 (document_230f106da00218449171.txt:p-0036); occ-0008 (document_230f106da00218449171.txt:p-0036)	Biological insight:Yes Evidence sufficiency:Sufficient Supporting quote + locator: To study the impact of batch effects on downstream performance, we employed scGen [53] for batch effect removal, leveraging its variational autoencoder (VAE) framework to learn a latent representation that disentangles biological signals from technical variation. Notes:
SCGEN-HV-026 FORMAL VALIDATION	The Power of Two: integrating deep diffusion models and variational autoencoders for single-cell transcriptomics analysis DOI: 10.1101/2023.04.13.536789 Article page: https://doi.org/10.1101/2023.04.13.536789 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2023/04/16/2023.04.13.536789.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_e6092864959832ed9def.txt:p-0024); occ-0002 (document_e6092864959832ed9def.txt:p-0024); occ-0003 (document_e6092864959832ed9def.txt:p-0024); occ-0004 (document_e6092864959832ed9def.txt:p-0024); occ-0005 (document_e6092864959832ed9def.txt:p-0026); occ-0006 (document_e6092864959832ed9def.txt:p-0026); occ-0007 (document_e6092864959832ed9def.txt:p-0026); occ-0008 (document_e6092864959832ed9def.txt:p-0058); occ-0009 (document_e6092864959832ed9def.txt:p-0058); occ-0010 (document_e6092864959832ed9def.txt:p-0058); occ-0011 (document_e6092864959832ed9def.txt:p-0058); occ-0012 (document_e6092864959832ed9def.txt:p-0060); occ-0013 (document_e6092864959832ed9def.txt:p-0062)	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use:Benchmark Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: We extend this analysis to other cell types and showed scVAEDer outperforms scGEN in predicting the effects of perturbation across different cell types (Fig. 4b). Notes:

SCGEN-HV-038 FORMAL VALIDATION	The manatee variational autoencoder model for predicting gene expression alterations caused by transcription factor perturbations DOI: 10.1038/s41598-024-62620-z Article page: https://doi.org/10.1038/s41598-024-62620-z Full text 1: https://www.nature.com/articles/s41598-024-62620-z.pdf Controller-registered target occurrence locator(s): occ-0001 (document_4bee3cf5dc371f4e90dd.txt:p-0001); occ-0002 (document_4bee3cf5dc371f4e90dd.txt:p-0006)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: scGen¹ VAE and Ghahramani et al.² generative adversarial network models exploit the latent space arithmetics for perturbation analyses. Notes:
SCGEN-HV-015 FORMAL VALIDATION	A practical guide to sequencing in neuropsychiatric research DOI: 10.1038/s44277-025-00041-0 Article page: https://doi.org/10.1038/s44277-025-00041-0 Full text 1: https://www.nature.com/articles/s44277-025-00041-0.pdf Controller-registered target occurrence locator(s): occ-0001 (document_93c07b9a77d02618c3c9.txt:p-0008)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: While various methods have been developed for DGE analysis in single-cell datasets [88,89,90,91], pseudobulking remains the standard and robustly validated approach [92,93,94].

		Evidence sufficiency:Sufficient Supporting quote + locator: To overcome this limitation of the fixed effects model, Lotfollahi et al. (2019) introduced the scGen method Notes:
SCGEN-HV-016 FORMAL VALIDATION	Neural network-assisted humanisation of COVID-19 hamster transcriptomic data reveals matching severity states in human disease DOI: 10.1016/j.ebiom.2024.105312 Article page: https://doi.org/10.1016/j.ebiom.2024.105312 Full text 1: https://edoc.mdc-berlin.de/id/eprint/24800/1/24800oa.pdf Controller-registered target occurrence locator(s): occ-0001 (document_a720c35675d8cb1f5507.txt:p-0003); occ-0002 (document_a720c35675d8cb1f5507.txt:p-0003); occ-0003 (document_a720c35675d8cb1f5507.txt:p-0003); occ-0004 (document_a720c35675d8cb1f5507.txt:p-0004); occ-0005 (document_a720c35675d8cb1f5507.txt:p-0016)	Review status:LOCKED Citation validity:Yes Method executed:Yes Uses (select all): Primary use:Apply to biological data Biological insight:Yes Evidence sufficiency:Sufficient Supporting quote + locator: Here, we employ an scGen-based framework to match COVID-19 disease severity levels across humans and hamster species, while also developing a metric for measuring interspecies similarity on a cellular level. Notes:
SCGEN-HV-032 FORMAL VALIDATION	Spatial single-cell mass spectrometry defines zonation of the hepatocyte proteome DOI: 10.1038/s41592-023-02007-6 Article page: https://doi.org/10.1038/s41592-023-02007-6 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2022/12/03/2022.12.03.518957.full.pdf Full text 2: https://www.nature.com/articles/s41592-023-02007-6.pdf	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Transfer learning might

	Controller-registered target occurrence locator(s): occ-0001 (document_3998cde33dc544a7a33e.txt:p-0030); occ-0002 (document_5f32cdfc8029a11592f1.txt:p-0007)	then extend the approach to many new areas, as already shown for single-cell transcriptomics data ²⁴ . Notes:
SCGEN-HV-013 FORMAL VALIDATION	Biologically Interpretable VAE with Supervision for Transcriptomics Data Under Ordinal Perturbations DOI: 10.1101/2024.03.28.587231 Article page: https://doi.org/10.1101/2024.03.28.587231 Full text 1: https://www.biorxiv.org/content/biorxiv/early/2024/03/29/2024.03.28.587231.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_a83cd4b670847c80df1a.txt:p-0008)	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all): Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Exploring and understanding how cellular responses evolve across a spectrum of stimulus doses sheds light on the underlying cellular mechanisms and paves the way for precision medicine approaches tailored to individualized dosedependent responses specifically in the context of disease and drug treatment (Bock et al., 2022; Lotfollahi et al., 2019). Notes:
SCGEN-HV-007 FORMAL VALIDATION	Discovering governing equations of biological systems through representation learning and sparse model discovery DOI: 10.1093/nargab/lqaf048 Article page: https://doi.org/10.1093/nargab/lqaf048	Review status: LOCKED Citation validity: Yes Method executed: No Uses (select all):

	Full text 1: https://www.biorxiv.org/content/biorxiv/early/2024/09/23/2024.09.19.613953.full.pdf Controller-registered target occurrence locator(s): occ-0001 (document_550c6cc33e6cb8e4f260.txt:p-0025)	Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: By carefully designing their architecture and loss function, autoencoders can help extract meaningful representations from complex biological data [18, 26–28]. Notes:
SCGEN-HV-023 FORMAL VALIDATION	Can Pharmaceutical World Models Become Scientific Instruments? A State-of-the-Art Review of Simulation, Reasoning, and Validation DOI: 10.51847/hn9ejvqjoc Article page: https://doi.org/10.51847/hn9ejvqjoc Full text 1: https://pharmacophorejournal.com/storage/files/article/8753dc7a-5010-4f82-ab26-e222c36d7a92-ujOPb8uV7cclw8pM/GdSs6Qs3wBWZkKW6.pdf Controller-registered target occurrence locator(s): occ-0001 (document_fc575441ba4c0efdc5f3.txt:p-0025)	Review status:LOCKED Citation validity:Yes Method executed:No Uses (select all): Primary use:Mention only Biological insight:No Evidence sufficiency:Sufficient Supporting quote + locator: Depending on the task, state may include molecular structure, target conformation, gene expression, cellular phenotype, tissue context, exposure, dose, prior treatment, and experimental conditions. Perturbation-response models demonstrate that latent cellular representations can be conditioned on interventions and used to predict changes across selected cellular contexts [12]. Notes: